Probabilistic Inventory models: Part 2

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Example (based on Operations Research by HA TAHA): An industry uses adhesive fluid in its manufacturing process at the rate of 1000 gallons per month. It costs Rs. 5000 to place an order. The holding cost per gallon per month is Rs. 20, and the shortage cost per gallon is Rs 100. Historical data show that the demand during lead time is uniformly distributed in the range (0, 100) gallons. Determine the optimal ordering policy.

Solution: Given,

D = 1000 gallons per month, K = Rs. 5000 per order, h = Rs. 20 per gallon per month

p = Rs. 100 per gallon, $f(x) = 1/100$, $0 \le x \le 100$, $E\{x\} = 50$ gallons

First step is to check whether the problem has a unique solution.

The model is based on three assumptions:

1. Unfilled demand during lead time is backlogged.
2. No more than one outstanding order is allowed.
3. The distribution of demand during lead time remains stationary with time.

To develop the total cost function per unit time, let

$f(x)$ = pdf of demand during lead time. Note: x is units of item.

D = Expected demand per unit time

h = Holding cost per inventory unit per unit time

p = Shortage cost per inventory unit

K = Setup cost per order

Using the equations for $\hat{y}$ and $\tilde{y}$

$$\hat{y} = \sqrt{\frac{2D\left(K + pE\{x\}\right)}{h}}$$

$$= \sqrt{\frac{2 \times 1000\left(5000 + 100 \times 50\right)}{20}}$$

$$= 1000$$

$$\tilde{y} = \frac{PD}{h}$$

$$= \frac{100 \times 1000}{20}$$

$$= 5000$$

As $\tilde{y} \geq \hat{y}$, a unique solution exists for y^* and R^*. S is computed as

$$S = \int_R^{100} (x - R) \frac{1}{100} \mathrm{d}x = \frac{R^2}{200} - R + 50$$

$$S = \int_R^{\infty} (x - R) f(x) \mathrm{d}x$$

The optimal values of y^* and R^* cannot be determined in closed forms.

The algorithm converges in a finite number of iterations, provided a feasible solution exists.

For $R = 0$, previous two equations yield

$$\hat{y} = \sqrt{\frac{2D(K + pS)}{h}}$$

$$\tilde{y} = \frac{PD}{h}$$

Unique optimal values of y and R exist when $\tilde{y} \geq \hat{y}$. The smallest value of y^* is

$$\sqrt{\frac{2KD}{h}}$$

It occurs when $S = 0$.

Using S in the following equation

$$y^* = \sqrt{\frac{2D\left(K + pS\right)}{h}}$$

y_i is obtained as

$$y_i = \sqrt{\frac{2D\left(K + pS\right)}{h}} \qquad (1)$$

$$= \sqrt{\frac{2 \times 1000\left(5000 + 100S\right)}{20}}$$

$$= \sqrt{500000 + 10000S}$$

Using the relation

one obtains

$$\int_{R^*}^{\infty} f(x)\,\mathrm{d}x = \frac{hy^*}{pD}$$

Solving,

$$\int_{R}^{100} \frac{1}{100}\,\mathrm{d}x = \frac{20y^*}{100\times 1000} \qquad (2)$$

$$R_i = 100 - \frac{y_i}{50}$$

Optimum solutions are determined by Eqs. (1) and (2).

Iteration 1:

$$y_1 = \sqrt{\frac{2KD}{h}} = \sqrt{\frac{2\times 5000\times 1000}{20}} = 707.16 \text{ gallons}$$

Iteration 2: $R_1 = 100 - \frac{y_1}{50} = 100 - \frac{707.16}{50} = 85.86 \text{ gallons}$

$$S = \frac{R_2^{\,2}}{200} - R_2 + 50 = \frac{85.86^2}{200} - 85.86 + 50 = 0.9997 \text{ gallons}$$

$$y_2 = \sqrt{500000 + 10000S} = \sqrt{500000 + 10000\times 0.9997} = 774.59 \text{ gallons}$$

Hence,

$$R_2 = 100 - \frac{y_2}{50} = 100 - \frac{774.59}{50} = 84.508 \text{ gallons}$$

Iteration 3:

$$S = \frac{R_2^{\,2}}{200} - R_2 + 50 = \frac{84.508^2}{200} - 84.508 + 50 = 1.20001 \,\text{gallons}$$

$$y_3 = \sqrt{500000 + 10000S} = \sqrt{500000 + 10000 \times 1.20001} = 715.542 \,\text{gallons}$$

Hence,

$$R_3 = 100 - \frac{y_3}{50} = 100 - \frac{715.542}{50} = 85.689 \,\text{gallons}$$

Solution:

Iteration 4:

$$S = \frac{R_3^{\,2}}{200} - R_3 + 50 = \frac{85.689^2}{200} - 85.689 + 50 = 1.02402 \text{ gallons}$$

$$y_4 = \sqrt{500000 + 10000S} = \sqrt{500000 + 10000 \times 1.02402} = 714.311 \text{ gallons}$$

Hence,

$$R_4 = 100 - \frac{y_4}{50} = 100 - \frac{714.311}{50} = 85.714 \text{ gallons}$$

Solution:

Iteration 5:

$$S = \frac{R_4^{\,2}}{200} - R_4 + 50 = \frac{85.714^2}{200} - 85.714 + 50 = 1.0204 \text{ gallons}$$

$$y_5 = \sqrt{500000 + 10000S} = \sqrt{500000 + 10000 \times 1.0204} = 714.2857 \text{ gallons}$$

Hence,

$$R_5 = 100 - \frac{y_5}{50} = 100 - \frac{714.2857}{50} = 85.714 \text{ gallons}$$

As $y_4 \approx y_5$ and $R_4 \approx R_5$, the optimum value is

R^* = 85.71 gallons and y^* is 714.29 gallons

No. of iterations	y_i (in gallons)	R_i (in gallons)
1	707.16	85.86
2	774.59	84.51
3	715.54	85.69
4	714.31	85.71
5	714.29	85.71

Single period models

- **Single-period models** in inventory management are used for decision-making when an item is ordered only once for a specific period.
- Unsold items cannot be carried over to the next period.
- At the end of period, leftover quantities are disposed off.
- The objective is to balance the cost of ordering too much (overstock) with the cost of ordering too little (stock out or lost sales).
- These models are typically applied in industries like **newspapers or seasonal goods**, where demand is uncertain and leftover inventory has little or no value.

Single-Period Model

- A single-period inventory model describes a situation in which one order is placed for a product. At the end of the sales period, any remaining product has little or no value.
  - Examples are seasonal goods, bakery goods, newspapers, and magazines.
- In other words, even though items are ordered weekly or daily, they cannot be held over and used as inventory in the next sales period.
- ***Therefore, the decision is to be made on how much to order at the beginning of the period.***

The symbols used in the development of the models include

K = Setup cost per order

h = Holding cost per held unit during the period

p = Penalty cost per shortage unit during the period

$f(D)$ = probability distribution function (pdf) of demand, D, during the period

y = Order quantity

x = Inventory on hand before an order is placed.

The model determines the optimal value of y that minimizes the sum of the expected holding and shortage costs. Given optimal y ($=y^*$), the inventory policy calls for ordering y^*-x if $x<y$; otherwise, no order is placed.

No-Setup Model (Newsvendor Model)

This model is known as the newsvendor model (the original classical name is the newsboy model). It deals with stocking and selling newspapers and periodicals.

The assumptions of the model are:

- Demand occurs instantaneously at the start of the period immediately after the order is received.
- No setup cost is incurred.

Characteristics of this model are as follows:

Single-period: Decisions are made for a single selling season or period.

Uncertain demand: Demand is not known with certainty in advance.

Limited shelf life: Inventory cannot be held over to the next period, meaning excess inventory has little or no value.

No-Setup Model (Newsvendor Model)

Figure demonstrates the inventory position after the demand, D, is satisfied. If $D < y$, the quantity $y–D$ is held during the period. Else, a shortage amount $D–y$ will result if $D > y$.

The expected cost for the period, $E\{C(y)\}$, is expressed as

$$E\{C(y)\} = h\int_0^y (y-D)f(D)\mathrm{d}D + p\int_y^\infty (D-y)f(D)\mathrm{d}D$$

D: expected demand per unit time
y: order quantity

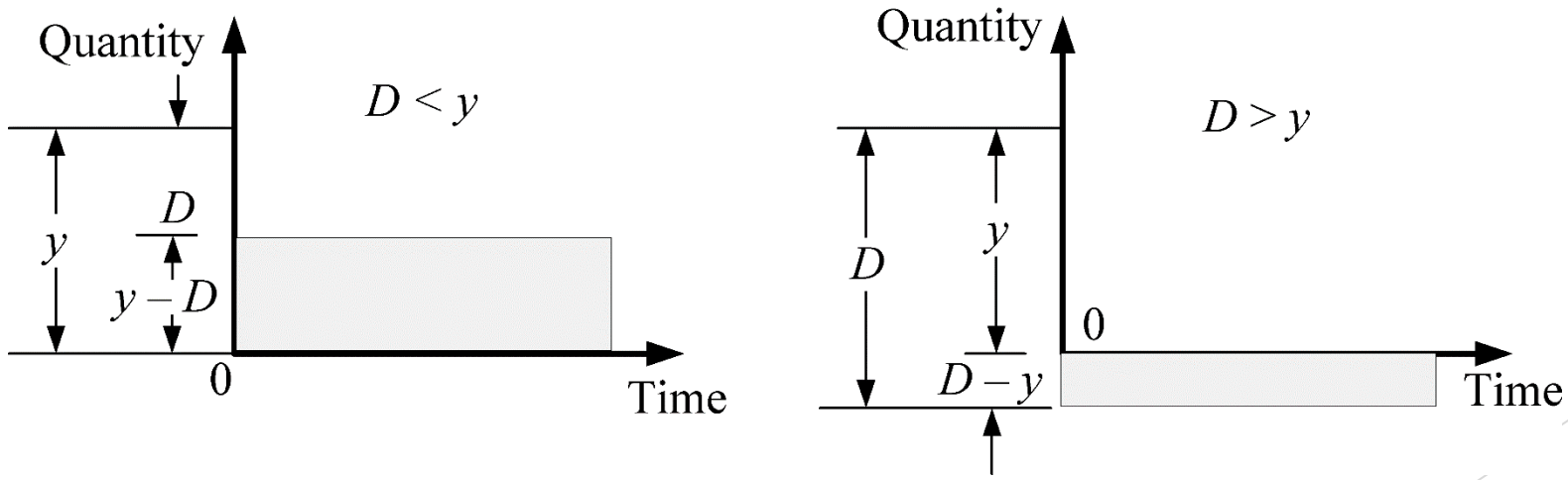


Figure: Shortage and holding inventory in single-period model

Single-Period Model

- The exact demand for such seasonal products is never known. Hence, a probability distribution related to demand is considered.
- If normal distribution is assumed, and the management stocked and sold an average (mean) of 100 Christmas trees each season, then there is a 50% chance that would stock out and a 50% chance that would have trees left over.
- To determine the optimal stocking policy for trees before the season begins, management need to know the standard deviation and consider two marginal costs: ***cost of shortage*** and ***cost of overage***

- C_s = Cost of shortage (underestimated)

  = Sales price per unit – Cost per unit
- C_o = Cost of overage (overestimated)

  = Cost per unit – Salvage value per unit (if there is any)
- The service level, the probability of not stocking out should be equal to or more than the ratio $\{C_s/(C_s + C_0)\}$

- ***Question:*** A newspaper seller sells 125 copies of a newspaper each day. He believes the sale of the newspaper is normally distributed, with a standard deviation of 15 papers. He pays Rs 3 for each paper and sells for Rs 5. He is given a Rs 1.5 credit for each unsold paper. He wants to determine how many papers he should order each day and the stock out risk for that quantity.

- ***Solution:***

C_s = Cost of shortage = Sales price per unit – Cost per unit

= Rs (5 – 3) = Rs 2

C_o = Cost of overage = Cost per unit – Salvage value per unit

= Rs (3 –1.5) = Rs 1.5

$$\frac{C_s}{C_s + C_0} = \frac{2}{2+1.5} = 0.571$$

- Now, the *Z* value has to be evaluated that yields a probability of 0.571. Using MS Excel command NORMSINV(0.571), the *Z* value is 0.1789.
- The command NORMSINV(p) in MS Excel returns the value z such that, with probability p, a standard normal random variable takes on a value that is less than or equal to z.
- Hence, optimal stocking level = 125 + 0.1789×Sandard Deviation

  = 125 + 0.1789×15

  = 128 newspapers

A derivation of single-period Model based on the book "Design and Analysis of Lean Production Systems" authored Ronald G. Askin and Jeffrey B. Goldberg, John Wiley & Sons, 2002

- c=purchase (production) cost of each item
- p=price per item sold (related to revenue)
- s= salvage value of each unsold item
- D=demand, a random variable with probability density function $f(x)$
- L=shortage cost for each unmet item demand
- Q= number of units ordered (the decision variable)

Expected Profit

- E(profit)=$(p-c)Q-L(D-Q)$, for $Q \le D$
- E(profit)=$pD-cQ+s(Q-D)$, for $Q>D$

$$E = p\int_0^Q xf(x)\mathrm{d}x + s\int_0^Q (Q-x)f(x)\mathrm{d}x + p\int_Q^\infty Qf(x)\,\mathrm{d}x - L\int_Q^\infty (x-Q)f(x)\mathrm{d}x - cQ$$

First term: revenue when demand is less than supply; Second term: salvage value
Third term: revenue when demand exceeds supply; Fourth term: shortage cost;
Fifth term: cost of goods

$$\frac{\mathrm{d}E}{\mathrm{d}Q}=\frac{\mathrm{d}}{\mathrm{d}Q}\left\{p\int_0^Q xf(x)\mathrm{d}x+s\int_0^Q\left(Q-x\right)f(x)\mathrm{d}x+p\int_Q^\infty Qf(x)\,\mathrm{d}x-L\int_Q^\infty\left(x-Q\right)f(x)\mathrm{d}x-cQ\right\}=0$$